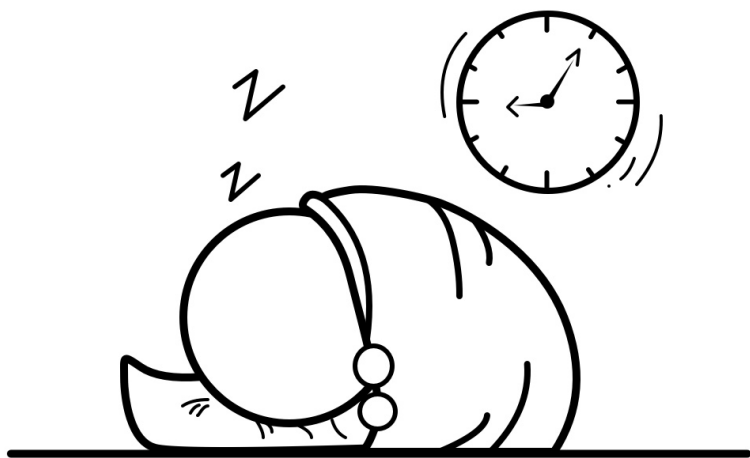


Time to Wake up

Daily Reminder



beautiful things happen
when you do the work to reprogram
that negative voice in your head

日期: 9 / 23

今天 - 2

$$1. (a) f(x) = x_1^2 + x_1 x_2 + x_2^2 - 3x_1 - 5x_2 \geq \frac{1}{2}(x_1^2 + x_2^2) - 3x_1 - 5x_2 \\ \geq \frac{1}{2}(x_1^2 + x_2^2) - 5|x_1 + x_2| \geq \frac{1}{2}(x_1^2 + x_2^2) - \sqrt{10}(x_1^2 + x_2^2)^{1/2} = \frac{1}{2}\|x\|^2 - \sqrt{10}\|x\|$$

$\therefore$ as $\|x\| \rightarrow +\infty$, $f(x) \rightarrow +\infty$. $\square$

(b) because $\|x\| \rightarrow +\infty$, $f(x) \rightarrow +\infty$, so $f(x)$ doesn't have maximum.

when $f(x)$ is minimum:

$$Df(x) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2} \right) = (2x_1 + x_2 - 3, x_1 + 2x_2 - 5) = \vec{0}$$

$$\text{so } \begin{cases} x_1 = \frac{1}{2} \\ x_2 = \frac{7}{2} \end{cases}, f(\vec{x})_{\min} = -\frac{19}{2}$$

$$2. (a) \text{ Don't. because } f(w) = \sum_{i=1}^m \log(1 + e^{-y_i x_i^T w}) > 0,$$

but use w_0 , s.t. $\forall i, y_i x_i^T w_0 > 0$,

$$\text{get } w = Nw_0, N \in \mathbb{N}^+, f(w) = \sum_{i=1}^m \log(1 + e^{-Ny_i x_i^T w_0})$$

let $N \rightarrow +\infty$, get $f(w) = \sum_{i=1}^m \log(1 + e^{-Ny_i x_i^T w_0}) \rightarrow 0$, so it doesn't have.

$$(b) i) \text{ if } \max_{1 \leq i \leq m} -y_i x_i^T w = -y_a x_a^T w.$$

$$R.) f(w) = \sum_{i=1}^m \log(1 + e^{-y_i x_i^T w}) \geq \log(1 + e^{-y_a x_a^T w})$$

$$\geq \log(e^{-y_a x_a^T w}) = -y_a x_a^T w. \text{ 得证!}$$

ii) because $\forall w, \exists i_0$, s.t. $y_{i_0} x_{i_0}^T w < 0$ ($-y_{i_0} x_{i_0}^T w > 0$)

$$\text{so } -y_a x_a^T w = \max_{1 \leq i \leq m} -y_i x_i^T w \geq -y_{i_0} x_{i_0}^T w > 0$$

$$\text{and because } |h(w)| = |y_i x_i^T w| = |x_i^T w| \leq \sum_{j=1}^n |x_{ij}| < +\infty$$

$\therefore h(w)$ is bounded, so $h(w)$ has global minimum.

日期:

/

$$\text{iii)} \quad h(\mathbf{w}) \geq h(\mathbf{w}_0) \|\mathbf{w}\|$$

$$\begin{aligned} h(\mathbf{w}) &= -\mathbf{y}_a \mathbf{x}_a^T \mathbf{w} = \|\mathbf{w}\| \cdot \mathbf{y}_a \mathbf{x}_a^T \frac{\mathbf{w}}{\|\mathbf{w}\|} \geq \theta_i \cdot \|\mathbf{w}\| - \mathbf{y}_i \mathbf{x}_i^T \frac{\mathbf{w}}{\|\mathbf{w}\|} \\ &= h(\mathbf{w}_0) \|\mathbf{w}\| \quad . \quad \square \end{aligned}$$

iv)

$$\text{c)} \quad f(\mathbf{w}) = \sum_{i=1}^m \log(1 + e^{-\mathbf{y}_i \mathbf{x}_i^T \mathbf{w}})$$

$$\therefore Df(\mathbf{w}) = \sum_{i=1}^m \frac{e^{-\mathbf{y}_i \mathbf{x}_i^T \mathbf{w}}}{1 + e^{-\mathbf{y}_i \mathbf{x}_i^T \mathbf{w}}} \cdot (-\mathbf{y}_i \mathbf{x}_i^T) = \sum_{i=1}^m -\frac{\mathbf{y}_i}{e^{\mathbf{y}_i \mathbf{x}_i^T \mathbf{w}} + 1} \mathbf{x}_i^T$$

$$\therefore \nabla f(\mathbf{w}) = (Df(\mathbf{w}))^T = \sum_{i=1}^m -\frac{\mathbf{y}_i}{e^{\mathbf{y}_i \mathbf{x}_i^T \mathbf{w}} + 1} \mathbf{x}_i$$

日期: /

$$3. (a) g(d) = f(x+d) - (f(x) + \nabla f(x)^T d)$$

$$h(d) = d^T d$$

$$\frac{g(d)}{h(d)} = \frac{g(d) - g(0)}{h(d) - h(0)} \stackrel{0 < t < 1}{=} \frac{g'(td)}{h'(td)} = \frac{d \nabla^2 f(x+td)}{2d} = \frac{\nabla^2 f(x+td)}{2}$$

$$\therefore g(d) = \frac{1}{2} d^T \nabla^2 f(x+td) d$$

$$\therefore f(x+d) = g(d) + f(x) + \nabla f(x)^T d = f(x) + \nabla f(x)^T d + \frac{1}{2} d^T \nabla^2 f(x+td) d$$

$$(b) \int_0^1 \nabla^2 f(x+td) d \, dt$$

$$= \int_0^1 D(\nabla f(x+td)) d \, (dt)$$

$$= \int_x^{x+d} D(\nabla f(x+td)) d(x+td)$$

$$= \nabla f(x+d) - \nabla f(x). \quad \square$$

4.

$$A = \begin{pmatrix} 7 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 3 \end{pmatrix} \quad \text{because } |7|, | \begin{smallmatrix} 7 & 2 \\ 2 & 4 \end{smallmatrix} |, | \begin{smallmatrix} 7 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 3 \end{smallmatrix} | > 0.$$

so A is positive definite.

$$B = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 2 & 1 \\ 0 & 1 & -3 \end{pmatrix} \quad \text{because } |1| > 0, | \begin{smallmatrix} 1 & 2 \\ 2 & 2 \end{smallmatrix} | < 0$$

so B is indefinite.

$$-C = \begin{pmatrix} 5 & -2 & -2 \\ -2 & 6 & 0 \\ -2 & 0 & 4 \end{pmatrix} \quad \text{because } |5| > 0, | \begin{smallmatrix} 5 & -2 \\ -2 & 6 \end{smallmatrix} | > 0, | \begin{smallmatrix} 5 & -2 & -2 \\ -2 & 6 & 0 \\ -2 & 0 & 4 \end{smallmatrix} | > 0$$

so C is negative definite.

日期: 10/11

第二题

1. Let $x_1, x_2 \in \mathbb{R}^n$, s.t. $f(x_1) = Ax_1 + b \in C$, $f(x_2) = Ax_2 + b \in C$.

$$2. f(\theta x_1 + \bar{\theta} x_2) = A(\theta x_1 + \bar{\theta} x_2) + b = \theta(Ax_1) + \bar{\theta}(Ax_2) + b.$$

$$= \theta(Ax_1 + b) + \bar{\theta}(Ax_2 + b)$$

because C is convex, so $\theta(Ax_1 + b) + \bar{\theta}(Ax_2 + b) \in C$.

$\therefore f(\theta x_1 + \bar{\theta} x_2) \in C$, so $f^{-1}(C)$ is convex!

2. Let $a_{1i}, a_{1j} \in C_1$, $a_{2i}, a_{2j} \in C_2$

$$x_1 = a_{1i} - a_{2i}, x_2 = a_{1j} - a_{2j}$$

$$\therefore \theta x_1 + \bar{\theta} x_2 = (\theta a_{1i} + \bar{\theta} a_{1j}) - (\theta a_{2i} + \bar{\theta} a_{2j})$$

because C_1, C_2 are convex, so $\theta a_{1i} + \bar{\theta} a_{1j} \in C_1$, $\theta a_{2i} + \bar{\theta} a_{2j} \in C_2$.

$\therefore \theta x_1 + \bar{\theta} x_2 = a_{1k} - a_{2k} \in C$, so C is convex!

3. (a) Let x_1, x_2 in $\text{int } C$. Then exist a ε , such that $U(x_1, \varepsilon), U(x_2, \varepsilon) \subset C$. So $x_3 = \theta x_1 + \bar{\theta} x_2$, consider $U(x_3, \varepsilon)$.

$U(x_3, \varepsilon) = x_3 + r\varepsilon, r \in [0, 1]$.

$$U(x_3, \varepsilon) = x_3 + r\varepsilon = \theta(x_1 + r\varepsilon) + \bar{\theta}(x_2 + r\varepsilon) \subset C$$

$$U(x_3, \varepsilon) = x_3 + r\varepsilon = \theta(x_1 + r\varepsilon) + \bar{\theta}(x_2 + r\varepsilon) \subset C$$

So $\text{int } C$ is a convex set.

(b) Let x, y in $\bar{C}$. so exists two array $\{x_k\}, \{y_k\}$, $x_k \in C, y_k \in C$

$$\lim_{k \rightarrow +\infty} x_k = x, \lim_{k \rightarrow +\infty} y_k = y.$$

$$\|\theta x_k + \bar{\theta} y_k - (\theta x + \bar{\theta} y)\| \leq \lambda \|x_k - x\| + (1-\lambda) \|y_k - y\|$$

so $\lim_{k \rightarrow +\infty} \|\theta x_k + \bar{\theta} y_k - (\theta x + \bar{\theta} y)\| = 0$, so $\theta x + \bar{\theta} y$ in $\bar{C}$.

日期: /

4. use the complete induction:

for $m=2$, $x_1, x_2 \in C$, so $\theta_1 x_1 + \theta_2 x_2 \in C$. Yes! (definition of convex set)

promise for $m=k$, the conclusion is true

$$\text{as for } m=k+1, \sum_{i=1}^{k+1} \theta_i x_i = \sum_{i=1}^k \theta_i x_i + \theta_{k+1} x_{k+1} = \sum_{i=1}^k \theta_i \underbrace{\sum_{j=1}^k \theta_j x_j}_{\in C} + \theta_{k+1} x_{k+1}$$

$$= (1 - \theta_{k+1}) \cdot x_{k+2} + \theta_{k+1} x_{k+1}, \text{ because } x_{k+2} \in C, x_{k+1} \in C$$

$$\text{so } (1 - \theta_{k+1}) x_{k+2} + \theta_{k+1} x_{k+1} \in C. \quad \square$$

$$5. \|x - x_0\|_2 \leq \|x - x_1\|_2 \Rightarrow (x - x_0)^T (x - x_0) \leq (x - x_1)^T (x - x_1)$$

$$\Rightarrow \underline{x^T x} - \underline{x^T x_0} - \underline{x_0^T x} + x_0^T x_0 \leq \underline{x^T x} - \underline{x^T x_1} - \underline{x_1^T x} + x_1^T x_1$$

$$\therefore (x_1^T - x_0^T) x + x^T (x_1 - x_0) \leq x_1^T x_1 - x_0^T x_0$$

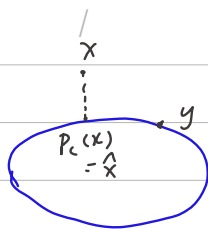
$$\therefore 2 (x_1^T - x_0^T) x \leq x_1^T x_1 - x_0^T x_0$$

$$\therefore A = \begin{pmatrix} 2(x_1^T - x_0^T) \\ \vdots \\ 2(x_n^T - x_0^T) \end{pmatrix}, \quad b = \begin{pmatrix} x_1^T x_1 - x_0^T x_0 \\ \vdots \\ x_n^T x_n - x_0^T x_0 \end{pmatrix}. \quad \square$$

日期:

第三次

1.



$$\hat{x} = p_C(x).$$

$$\text{证: } \|x - y\|^2 = \|(x - \hat{x}) - (y - \hat{x})\|^2$$

$$= (x - \hat{x})^2 + (y - \hat{x})^2 - 2(x - \hat{x})(y - \hat{x})$$

$$\geq (x - \hat{x})^2 + (y - \hat{x})^2 > (y - \hat{x})^2 = \|y - \hat{x}\|^2. \quad \square$$

2. 设 $x_1^*, x_2^* \in M$.

$$\text{证: } f(\theta x_1^* + \bar{\theta} x_2^*) \leq \theta f(x_1^*) + \bar{\theta} f(x_2^*) \leq \theta f(x) + \bar{\theta} f(x) = f(x).$$

$$\text{故 } \theta x_1^* + \bar{\theta} x_2^* \in M. \quad \square$$

$$3. (a) f(x) = x^T \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix} x,$$

$$\text{证: } f''(x) = 2 \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix} \triangleq A.$$

而 A 为半正定阵, 故 $f(x)$ 为凸函数.

$$(b) f''(x) = \begin{pmatrix} \frac{2}{x_1^3 x_2} & \frac{1}{x_1^2 x_2^2} \\ \frac{1}{x_1^2 x_2^2} & \frac{2}{x_1 x_2^3} \end{pmatrix}, \text{ 为正定矩阵}$$

故 $f(x)$ 为凸函数

$$(c) f''(x) = \begin{pmatrix} 0 & 2x_2 \\ 2x_2 & 2x_1 \end{pmatrix} \text{ 为不定矩阵, 故既不是凸也不是凹.}$$

日期: /

$$(d) f'(x) = (x_2^{-\frac{1}{2}}, -\frac{1}{2} x_2^{-\frac{3}{2}} x_1)$$

$$f''(x) = \begin{pmatrix} 0 & -\frac{1}{2} x_2^{-\frac{3}{2}} \\ -\frac{1}{2} x_1 x_2^{-\frac{3}{2}} & \frac{3}{4} x_2^{-\frac{5}{2}} x_1 \end{pmatrix} \text{ 也是不定矩阵, } f(x) \text{ 非凹非凸.}$$

$$(e) f''(x) = -\alpha(1-\alpha) x_1^{\alpha-2} x_2^{-\alpha-1} \begin{pmatrix} x_2^2 & -x_1 x_2 \\ -x_1 x_2 & x_1^2 \end{pmatrix} \text{ 是半负定矩阵.}$$

故 $f(x)$ 是凹函数.

4. $f(x) = \|x\|$ 是凸函数. 仿射函数 $\|Ax+b\|$ 为凸函数.

而 $h(x) = x^3$ 在 $x > 0$ 时单增凸, $g(x) = \|Ax+b\|$ 在 $\mathbb{R}$ 上凸
 $\Rightarrow h(g(x)) = \|Ax+b\|^3$ 在 $\mathbb{R}$ 上凸.

又函数 $\log(1+e^{x_1+x_2})$ 在 $\mathbb{R}^2$ 上凸, 经过 $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ 仿射变换,

$\log(1+e^{2x_1+3x_2})$ 凸. 而它的下水平集是凸集, 故 $\log(1+e^{2x_1+3x_2}) \leq 2$ 凸.

$\therefore \max \{ \|Ax+b\|^3, \log(1+e^{2x_1+3x_2}) \leq 2 \}$ 为凸函数.

5.

$$(1) \Rightarrow "f(y) \geq f(x) + \langle \nabla f(x), y-x \rangle \quad (1)$$

$$f(x) \geq f(y) + \langle \nabla f(y), x-y \rangle \quad (2)$$

$$(1) + (2) \text{ 得 } \langle \nabla f(x) - \nabla f(y), x-y \rangle \geq 0. \quad \square$$

(2) " $\Leftarrow$ " 见反:

日期:

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设 $g(t) = f(x + td)$

$$\therefore g'(t) = f'(x + td)d \quad \forall t_1, t_2$$

令 $x = x + t_1d, y = x + t_2d$. 有 $\langle \nabla f(x + t_1d) - \nabla f(x + t_2d), (t_1 - t_2)d \rangle \geq 0$

$$(g'(t_1) - g'(t_2))(t_1 - t_2) = (f'(x + t_1d)d - f'(x + t_2d)d)(t_1 - t_2)$$

$$= \langle \nabla f(x + t_1d) - \nabla f(x + t_2d), (t_1 - t_2)d \rangle \geq 0$$

$\therefore \forall t \forall x, d; g(t) = f(x + td)$ 为凸函数 $\Rightarrow f(x)$ 为凸函数.

日期:

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第四次

(a) $x_1 - 3x_2 = 0 \Leftrightarrow \begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$ 为 Affine function

而由 x^2 凸, e^x 凸, $\Rightarrow f(x) = x^2 + e^x$ 凸.

则 $(x_1 - x_2)^2 + 4x_1x_2 + e^{x_1+x_2} = (x_1+x_2)^2 + e^{x_1+x_2}$ 凸,

而 $(x_1+x_2)^2 + e^{x_1+x_2} \leq 0$ 为 sublevel set (其实是空集) $\Rightarrow$ 凸.

而 $g(x_1, x_2) = 2x_1^2 - 2x_1x_2 + x_2^2 + x_1 + x_2$,

$\nabla g = (4x_1 - 2x_2 + 1, -2x_1 + 2x_2 + 1)$,

$\nabla^2 g = \begin{pmatrix} 4 & -2 \\ -2 & 2 \end{pmatrix} \succ 0$. 则也为凸函数.

则 a 为凸优化问题.

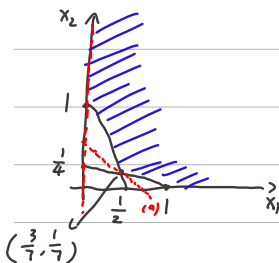
(b) x_1^2 凸, x_2^4 凸, $x_1^2 + 3x_2^4$ 凸

但 $5x_1^2 - 7x_2 = 0$ 不是 Affine 也不凸, 则 (b) 不是凸优化问题.

日期: /

2. feasible set:

蓝笔划出的即为可行域。



$$(a) (x_1 + x_2)_{\min} = \frac{3}{7} + \frac{1}{7} = \frac{4}{7}$$

$$(b) (x_1 - x_2)_{\min} = -\infty, \text{无下限}$$

$$(c) (\lambda_1)_{\min} = 0$$

(d) 输出结果

```
warnings.warn(ECOS_DEPRECATION_MSG, FutureWarning)
status: optimal
optimal value 0.3333333331727654
optimal var 0.3333333331727654 0.3333333329351904
```

(e)

```
status: optimal
optimal value 0.346938775510204
optimal var 0.42857142857142855 0.14285714285714285
```

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3. (a) $\|x\|_\infty \leq 1 \Leftrightarrow \forall i, -1 \leq x_i \leq 1$

令 $y = \|Ax - b\|_1$, 则目标函数 $\Rightarrow \min_x^T y$.

限制条件加上 $-y \leq Ax - b \leq y$.

即为 $\min_{x,y}$
 $s.t. \quad -y \leq Ax - b \leq y$
 $-1 \leq x \leq 1$

$\|3x_1 + x_2 - 5\| \leq y_1$
 $\|x_1 + 3x_2 - 8\| \leq y_2$
 $\|x_1 + 2x_2 + 5\| \leq y_3$

(b) $A = \begin{pmatrix} 3 & 1 \\ 1 & -3 \\ 1 & 2 \end{pmatrix}$ (1, -1)
 $b = \begin{pmatrix} 5 \\ 8 \\ -5 \end{pmatrix}$
 python 代码 $[3, 1, 1], [1, -3, 2]$

$\min(y_1 + y_2 + y_3) \begin{pmatrix} -3 \\ -4 \\ -4 \end{pmatrix}$

输出

```
Optimal value 10.999999998676984
Optimal var
[ 1. -1.]
```

(c) $\begin{cases} Ax - y \leq b \\ -Ax - y \leq -b \end{cases}$

$\begin{pmatrix} 3 & 1 & -1 & 0 & 0 \\ 1 & -3 & 0 & -1 & 0 \\ 1 & 2 & 0 & 0 & -1 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} \leq b$ 目标: $[0, 0, 1, 1, 1]$ $\begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix}$

$\begin{pmatrix} -3 & -1 & -1 & 0 & 0 \\ -1 & 3 & 0 & -1 & 0 \\ -1 & -2 & 0 & 0 & -1 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} \leq -b$

$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} \leq \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$

输出,

```
The optimal value is 10.999999991758674
the Optimal Var is [ 1. -1. 3. 4. 4.]
```

日期: /

4.

$$(a) w = (X^T X)^{-1} X^T y = \begin{pmatrix} \frac{3}{2} \\ 2 \end{pmatrix}$$

$$(b) X = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad y = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$$

$$\min \|Xw - y\|_2^2 \quad \text{s.t. } \|w\|_1 \leq t$$

① $t = 1$:

```
status: optimal
optimal value 9.0000000633334
optimal var [9.99962136e-01 3.78500362e-05]
```

与(a)不一样

② $t = 0$:

```
status: optimal
optimal value 4.000000000012024
optimal var [1.49999883 1.99999744]
```

与(a)解不一样。

(c) ① $t = 1$

```
status: optimal
optimal value 7.857489685034338
optimal var [0.86270479 0.50570788]
```

与(a)解不一样

② $t = 100$

```
status: optimal
optimal value 4.0000000000000195
optimal var [1.50000002 2.00000013]
```

与(a)解一样。

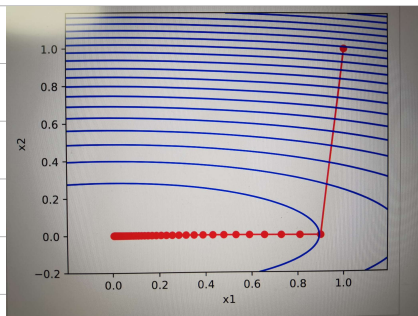
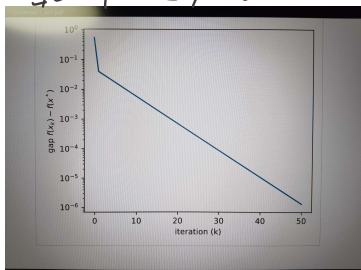
日期: /

5.

$$(a) L = \lambda_{\max}(Q) = \max(r, 1)$$

$$(b) Q = \begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}, \quad gd = Qx$$

stepsize = 1/77:



当step=2-2A时会溢出。

T6

$$\min \|xw - y\|_2^2$$

$$= (xw - y)^T (xw - y) = w^T x^T x w - w^T x^T y - y^T x w + y^T y$$

$$\therefore \nabla f(w) = 2x^T x w - 2x^T y$$

step-size = 0.01, epoch=1000. 有结果如下.

```
w = np.array([0.0,0.0]) #初始向量
w_traces = []
for i in range(1000):
    w_traces.append(w.copy()) #复制
    w_grad = fp(w)
    if(np.linalg.norm(w_grad) < 1e-5):
        break
    w -= w_grad * 0.01
print(w)
print(w_traces)
```

✓ 0.0s

[1.5 1.99999505]

日期: /

T7

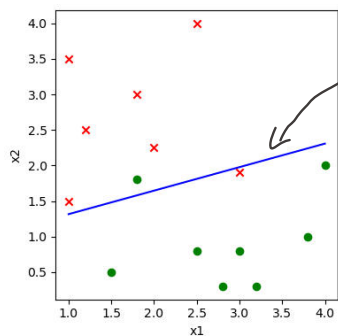
$$f(w) = \sum_{i=1}^m \log(1 + e^{-y_i x_i^T w})$$

$$\nabla f(w) = - \sum_{i=1}^m [1 - \sigma(y_i x_i^T w)] y_i x_i$$

$$\text{accuracy} = \frac{1}{m} \sum_{i=1}^m \mathbb{1}\{y_i = \hat{y}_i\}$$

$$\text{sig}(z) = \frac{1}{1 + e^{-z}}$$

$$\text{accuracy} = 0.8666666666666667$$



$$w = [-1.4702573, 4.44147843, -4.37142527]$$

日期:

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第五次

1.1 Suppose $f(\mathbf{x})$ is differentiable and α -strongly convex, and $g(\mathbf{x})$ is β -smooth. Show that the function $h(\mathbf{x}) = f(\mathbf{x}) - g(\mathbf{x})$ is convex if $\alpha \geq \beta$.

解: 令 $\tilde{f}(\mathbf{x}) = f(\mathbf{x}) - \frac{\alpha}{2} \|\mathbf{x}\|^2$, 其中 $\tilde{f}(\mathbf{x})$ 为凸函数.

$$\text{则 } h(\mathbf{x}) = \tilde{f}(\mathbf{x}) + \frac{\alpha}{2} \|\mathbf{x}\|^2 - g(\mathbf{x})$$

$$\therefore \nabla^2 h(\mathbf{x}) = \nabla^2 \tilde{f}(\mathbf{x}) + \alpha \mathbf{I} - \nabla^2 g(\mathbf{x})$$

由于 $\nabla^2 \tilde{f}(\mathbf{x}) \succeq 0$ ($\tilde{f}(\mathbf{x})$ 凸), 考虑 $A = \alpha \mathbf{I} - \nabla^2 g(\mathbf{x})$ 的特征值.

由 $g(\mathbf{x})$ β -smooth 有 $|\lambda(\nabla^2 g(\mathbf{x}))| \leq \beta$

而 $\alpha \geq \beta$, 故 $\lambda(A) = \alpha - \lambda(\nabla^2 g(\mathbf{x})) \geq \alpha - \beta \geq 0$

$\therefore A \succeq 0$, 故 $h(\mathbf{x})$ 为凸函数.

□

2. Consider the following optimization problem of Boyd and Vandenberghe,

$$\min_{x_1, x_2} f(x_1, x_2) = e^{x_1 + 3x_2 - 0.1} + e^{x_1 - 3x_2 - 0.1} + e^{-x_1 - 0.1} \quad (1)$$

In this problem, you will implement gradient descent with backtracking line search. First complete the function `gd_armijo` in `gd.py`. For problem (f) and (g), first implement the **pure** Newton's method in `newton.py`.

(a). What's the optimal solution x^* and the optimal value $f(x^*)$?

$$(a) \nabla f = (e^{x_1 + 3x_2 - 0.1} + e^{x_1 - 3x_2 - 0.1} - e^{-x_1 - 0.1}, 3e^{x_1 + 3x_2 - 0.1} - 3e^{x_1 - 3x_2 - 0.1}) = 0$$

$$(b) (x_1, x_2) = (-\frac{1}{2} \log 2, 0), f(x_1, x_2) = 2\sqrt{2}e^{-0.1} = 2.55926$$

(b) Gradient descent with backtracking line search

```
1: initialization  $x \leftarrow x_0 \in \mathbb{R}^n$ 
2: while  $\|\nabla f(x)\| > \delta$  do
3:    $t \leftarrow t_0$ 
4:   while  $f(x - t\nabla f(x)) > f(x) - \alpha\|\nabla f(x)\|_2^2$  do
5:      $t \leftarrow \beta t$ 
6:   end while
7:    $x \leftarrow x - t\nabla f(x)$ 
8: end while
9: return  $x$ 
```

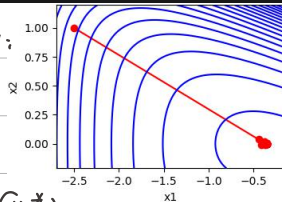
← the process of backtracking line search

$$(1) x_0 = (-2.5, 1)$$

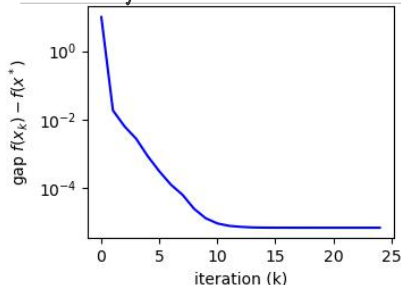
1) solution & outer iterations & inner iterations:

```
gradient descent with Armijo
number of iterations in outer loop: 24
total number of iterations in inner loop: 90
solution: [-3.46574844e-01 -3.90088025e-07]
value: 2.559266696661104
```

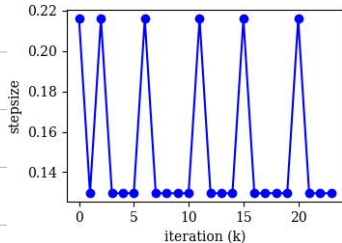
2) trajectory:



3) error $f(x_k) - f(x^*)$:



4) stepsize t_k :



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$$\textcircled{2} x_0 = (2.5, 1)$$

1) solution & outer iterations & inner iterations:

gradient descent with Armijo

number of iterations in outer loop: 38

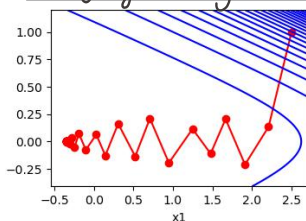
total number of iterations in inner loop: 181

solution: $[-3.46572426e-01 \ -4.21102179e-07]$

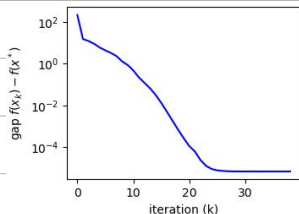
value: 2.5592666966609707

(has more loops than $(-2.5, 1.0)$)

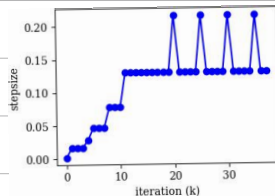
2) trajectory:



3) error $f(x_k) - f(x^*)$



4) stepsize t_k



(c) solution & iteration:

gradient descent with constant stepsize 0.1

number of iterations: 44

solution: $[-3.46576950e-01 \ -1.17422482e-17]$

value: 2.559266696672658

(d) ① size = 0.005;

② size = 0.1

gradient descent with constant stepsize 0.005

number of iterations: 10000

solution: [nan nan]

value: nan

gradient descent with constant stepsize 0.1

number of iterations: 100000

solution: [nan nan]

value: nan

=> Whether it's converge $\Leftarrow$ the initial point x_0

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(e) Newton's method (cont'd)

⇐ process of newton's method

- 1: initialization $x \leftarrow x_0 \in \mathbb{R}^n$
- 2: **while** $\|\nabla f(x)\| > \delta$ **do**
- 3: $x \leftarrow x - [\nabla^2 f(x)]^{-1} \nabla f(x)$
- 4: **end while**
- 5: **return** x

stop

$$\nabla f(x) = \begin{pmatrix} e^{x_1+3x_2-0.1} + e^{x_1-3x_2-0.1} - x_1-0.1, & 3e^{x_1+3x_2-0.1} - 3e^{x_1-3x_2-0.1} \end{pmatrix},$$

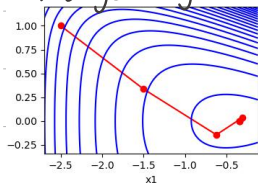
$$\nabla^2 f(x) = \begin{pmatrix} e^{x_1+3x_2-0.1} + e^{x_1-3x_2-0.1} - x_1-0.1, & 3e^{x_1+3x_2-0.1} - 3e^{x_1-3x_2-0.1} \\ 3e^{x_1+3x_2-0.1} - 3e^{x_1-3x_2-0.1}, & 9e^{x_1+3x_2-0.1} + 9e^{x_1-3x_2-0.1} \end{pmatrix}$$

① $x_0 = (-2.5, 1)$:

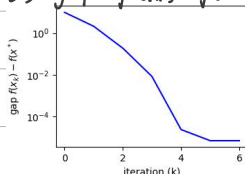
1) solution & iteration:

```
Newton's method
number of iterations: 6
solution: [-3.46573590e-01  9.31588223e-12]
value: 2.5592666966582156
```

2) trajectory:



3) gap $f(x_k) - f(x^*)$:

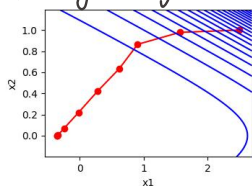


② $x_0 = (2.5, 1.0)$:

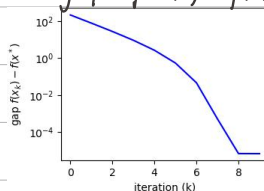
1) solution & iteration:

```
Newton's method
number of iterations: 9
solution: [-3.46573573e-01  1.13833135e-08]
value: 2.5592666966582165
```

2) trajectory:



3) gap $f(x_k) - f(x^*)$:



/

w.shape = (3, 1)

Damped Newton's method

```

1: initialization  $x \leftarrow x_0 \in \mathbb{R}^n$ 
2: while  $\|\nabla f(x)\| > \delta$  do
3:    $d \leftarrow -[\nabla^2 f(x)]^{-1} \nabla f(x)$  ▷ SC
4:    $t \leftarrow 1$ 
5:   while  $f(x + td) > f(x) + \alpha t \nabla f(x)^T d$  do
6:      $t \leftarrow \beta t$ 
7:   end while
8:    $x \leftarrow x + td$ 
9: end while
10: return  $x$ 

```

where $\alpha, \beta \in (0, 1)$

$$\nabla f(\mathbf{w}) = - \sum_{i=1}^m [1 - \sigma(y_i \mathbf{x}_i^T \mathbf{w})] y_i \mathbf{x}_i$$

最后全是！

$$\nabla^2 f(w) = \sum_{i=1}^m \sigma'(y_i x_i^T w) x_i x_i^T$$

$$\nabla^2 f(w) = - \sum_{i=1}^m [-\sigma'(y_i x_i^T w)] \cdot y_i^* x_i x_i^T$$

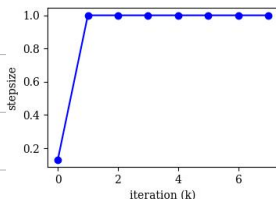
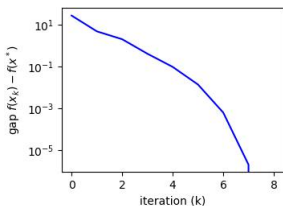
$$= \sum_{i=1}^m \sigma'(y_i x_i^T w, x_i x_i^T)$$

(b) ^①solution & iterations

```
number of iterations in outer loop: 8
total number of iterations in inner loop: 0
solution: [-1.4702126  4.44400571 -4.37587394]
value: 2.8766810999991255
```

② gap $f(x_k) - f(x^*)$.

③ stepsize τ_k .



(c) 用纯牛顿解法, 会导致矩阵求逆错误.

```
File "C:\array_function__internals.py", line 200, in inv
File "D:\rayfiles\Anaconda\Lib\site-packages\numpy\linalg\linalg.py", line 538, in inv
    ainv = _umath_linalg.inv(a, signature=signature, extobj=extobj)
                                ~~~~~^~~~~~
File "D:\rayfiles\Anaconda\Lib\site-packages\numpy\linalg\linalg.py", line 89, in _raise_linalgerror_singular
    raise LinAlgError("Singular matrix")
numpy.linalg.LinAlgError: Singular matrix
```

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4. Consider the optimization problem $\min_x f(x)$, where $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = (x-a)^4$, and $a \in \mathbb{R}$ is a constant.

(a). Find an explicit expression for the Newton step.

(b). Let x_k be the sequence of iterates generated by Newton's method. Let $y_k = |x_k - a|$ be the error between the k -th iterate and the optimal solution. Show that $y_{k+1} = \frac{2}{3}y_k$.

(c). Conclude $|x_k - a|$ decays to zero exponentially, i.e. x_k converges exponentially to a .

$$(a) \nabla f(x) = 4(x-a)^3, \quad (\nabla^2 f(x))^{-1} = \frac{1}{12(x-a)^2}$$

$$\therefore x_{k+1} = x_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k) = x_k - \frac{x_k - a}{3} = \frac{2}{3}x_k + \frac{1}{3}a$$

$$(b) |x_{k+1} - a| = \frac{2}{3}x_k - \frac{2}{3}a = \frac{2}{3}|x_k - a|$$

$$\therefore y_{k+1} = \frac{2}{3}y_k$$

$$(c) \text{ 由 } y_{k+1} = \frac{2}{3}y_k, \text{ 知 } y_n = \left(\frac{2}{3}\right)^{n-1}y_1$$

$$\therefore \lim_{n \rightarrow \infty} y_n = 0, \quad \text{即 } \lim_{n \rightarrow \infty} |x_n - a| = 0, \quad \text{即 } \lim_{n \rightarrow \infty} x_n = a. \quad \square$$

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$$X = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{matrix} 3 \times 2 \\ 2 \times 1 \end{matrix} \quad \begin{matrix} 2 \times 1 \\ 2 \times 1 \end{matrix} \quad \begin{matrix} 2 \times 3 \\ 3 \times 1 \end{matrix}$$

5. Complete the implementation of the soft-thresholding operator and ISTA in `ista.py` for solving Lasso in penalized form,

$$f(w) = \frac{1}{2} \|Xw - y\|_2^2 + \lambda \|w\|_1$$

We will explore the effect of λ on the number of zeros in the solution w^* .

for:

$$\text{prox}_h(y) = \underset{x}{\text{argmin}} \left\{ \frac{1}{2} \|x - y\|_2^2 + h(x) \right\}$$

so

$$x_{k+1} = \text{prox}_{t_k h}(x_k - t_k \nabla g(x_k))$$

Proximal gradient descent with constant step size

- 1: initialization $x \leftarrow x_0 \in \mathbb{R}^n$
- 2: **while** stopping criterion not satisfied **do**
- 3: $x \leftarrow \text{prox}_{t_k h}(x - t_k \nabla g(x))$
- 4: **end while**
- 5: **return** x

$$f(w) = \frac{1}{2} \|Xw - y\|_2^2$$

$$\nabla f_i(w) = X^T(Xw - y)$$

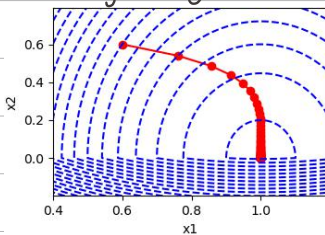
$$\therefore u_k - t_k \nabla f(u_k) = u_k - t_k X^T(Xu_k - y)$$

$$\text{prox}_{\lambda t}(w_i) = \begin{cases} w_i - \lambda t & w_i > \lambda t \\ 0 & |w_i| < \lambda t \\ w_i + \lambda t & w_i < -\lambda t \end{cases} = \text{sign}(w_i) \max(|w_i| - \lambda t, 0)$$

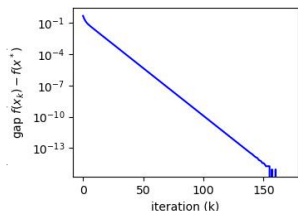
(a) ① solution & iteration

```
lambda = 2
number of iterations: 170
solution: [1.00000000e+00 9.98568486e-09]
value: 6.5
```

③ trajectory:



③ gap:



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(b)

```
lambda = 1
number of iterations: 167
solution: [1.25      0.99999999]
value: 4.875
```

there are no zeros.

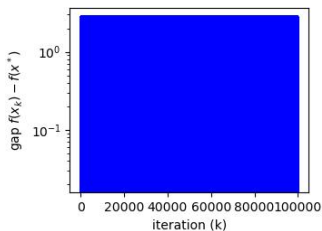
(c)

```
lambda = 7
number of iterations: 100000
solution: [-0.25      -0.40448048]
value: 15.59712658395166
```

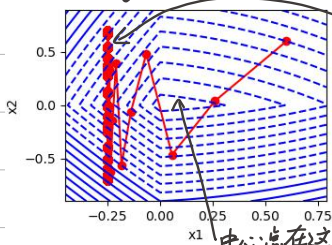
同样没有0?

但观察error曲线, 发现其非常不稳定:

① gap:



② trajectory:



中心点在这, 但它似乎在振荡.

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第六次

1. Consider the equality constrained quadratic program

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) = x_1^2 + x_1 x_2 + x_2^2 - 2x_1 - 3x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 = 1 \end{aligned}$$

(a). Find the optimal solution x_1^*, x_2^* by reduction to an unconstrained problem.(b). Find the optimal solution and the corresponding Lagrange multiplier λ^* using the Lagrangian multipliers method.

$$\begin{aligned} (a) \quad f(x_1, x_2) &= x_1^2 + x_1 x_2 + x_2^2 - 2x_1 - 3x_2 = (1-2x_2)^2 + (1-2x_2)x_2 + x_2^2 - 3x_2 - 2(1-2x_2) \\ &= 3x_2^2 - 2x_2 - 1 = 3\left(x_2 - \frac{1}{3}\right)^2 - \frac{4}{3} \\ \Rightarrow \quad &\begin{cases} x_1^* = \frac{1}{3} \\ x_2^* = \frac{1}{3} \end{cases} \end{aligned}$$

$$(b). \quad F(x_1, x_2, \lambda) = x_1^2 + x_1 x_2 + x_2^2 - 2x_1 - 3x_2 + \lambda(x_1 + 2x_2 - 1)$$

$$\therefore \begin{cases} \frac{\partial F}{\partial x_1} = 0 \\ \frac{\partial F}{\partial x_2} = 0 \\ x_1 + 2x_2 = 1 \end{cases} \Rightarrow \begin{cases} 2x_1^* + x_2^* + \lambda^* = 2 \\ x_1^* + 2x_2^* + 2\lambda^* = 3 \\ x_1^* + 2x_2^* = 1 \end{cases} \Rightarrow \begin{cases} x_1^* = \frac{1}{3} \\ x_2^* = \frac{1}{3} \\ \lambda^* = 1 \end{cases}$$

2. Solve the following problem,

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) = x_1 x_2 \\ \text{s.t.} \quad & x_1^2 + 2x_2^2 = 1 \end{aligned}$$

$$F(x_1, x_2, \lambda) = x_1 x_2 + \lambda(x_1^2 + 2x_2^2 - 1)$$

$$\therefore \begin{cases} \frac{\partial F}{\partial x_1} = 0 \\ \frac{\partial F}{\partial x_2} = 0 \\ x_1^2 + 2x_2^2 = 1 \end{cases} \Rightarrow \begin{cases} x_2^* + 2\lambda x_1^* = 0 \\ x_1^* + 4\lambda x_2^* = 0 \\ x_1^{*2} + 2x_2^{*2} = 1 \end{cases} \Rightarrow \begin{cases} x_1^* = \pm \frac{\sqrt{2}}{2} \\ x_2^* = \mp \frac{1}{2} \\ \lambda = \frac{\sqrt{2}}{4} \end{cases} \quad \text{或} \quad \begin{cases} x_1^* = \pm \frac{\sqrt{2}}{2} \\ x_2^* = \pm \frac{1}{2} \\ \lambda = -\frac{\sqrt{2}}{4} \end{cases}$$

$\downarrow$ 全局最小 $\downarrow$ 全局最大

$$\therefore x_1^* = \pm \frac{\sqrt{2}}{2}, x_2^* = \mp \frac{1}{2}, \min f = -\frac{\sqrt{2}}{4}$$

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T₃

$$(a) F = X^T A X + \mu (X^T X - 1)$$

$$\therefore \begin{cases} X^T X = 1 \\ \frac{\partial F}{\partial X} = 2AX + 2\mu X = 0 \end{cases} \Rightarrow \begin{cases} X^T X = 1 \\ AX = -\mu X \end{cases}$$

$$\text{求 } F \text{ 的最小值 } F_{\min} = X^T A X = X^T (-\mu X) = -\mu X^T X = -\mu,$$

$$\text{则 } -\mu \text{ 要最小} \Rightarrow -\mu = \lambda_1, \text{ 则 } AX = \lambda_1 X \text{ 且 } F_{\min} = \lambda_1. \quad \square$$

$$(b) i) \quad X^T A X + \mu_1 (X^T X - 1) + \mu_2 V_1^T X = 0$$

$$g_1(X) = X^T X - 1, \quad g_2(X) = V_1^T X$$

$\nabla g_1(X) = 2X$, $\nabla g_2(X) = V_1$, 当 $X \neq kV_1$ 时线性无关, 而当 $X = kV_1$ 时, $V_1 \cdot V_1 = 0$. 又 $V_1 = X = 0$, 矛盾!

$\therefore$ 由 Lagrange, 有 $0 = 2AX^* + 2\mu_1 X^* + \mu_2 V_1$, 则 $C_0 = -\mu_1$, $C_1 = -\frac{1}{2}\mu_2$.

ii) $AX^* = C_0 X^* + C_1 V_1$, 两边乘 V_1^T , 有

$$V_1^T A X^* = C_0 V_1^T X^* + C_1 V_1^T V_1 = C_1 V_1^T V_1 = 0 \quad \text{①}$$

$$\text{而 } V_1^T A X^* = (V_1^T A) X^* = \lambda_1 V_1^T X^* = 0 \dots \text{②, 由 ① ② 知 } C_1 = 0.$$

iii) 由 ii), 有 $AX^* = C_0 X^*$, 则 λf 有 $X^{*T} A X^* = C_0 \|X^*\|_2^2 = C_0$.

而 C_0 必须最小.

$$\text{① } C_0 = \lambda_1, AX^* = \lambda_1 X^*. \text{ 由于 } V_1^T X_1^* = 0, V_1 \perp X_1^*$$

由奇实, $\exists i (2 \leq i \leq n)$, $AX^* = \lambda_i X^*$, 而 $AX^* = \lambda_1 X^*$, 矛盾! (除非 $\lambda_1 = \lambda_2$)

② $C_0 = \lambda_2$, 可以成立, 取上文 $i=2$ 即可. 故最小值 $= C_0 = \lambda_2. \quad \square$

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T4

4. Consider the following problem,

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & f(x) = x_1^2 + x_2^2 \\ \text{s.t.} \quad & g_1(x) = (x_1 + 1)^2 + (x_2 - 1)^2 - 1 \leq 0 \\ & g_2(x) = x_1^2 + (x_2 - 1)^2 - 1 \leq 0 \end{aligned}$$

Write down the KKT conditions and find the optimal point x^* and the corresponding Lagrange multipliers.

$$\begin{cases} \lambda_1 (2x_1, 2x_2) + \mu_1 (2(x_1+1), 2(x_2-1)) + \mu_2 (2x_1, 2(x_2-1)) = 0 \\ \mu_1 \geq 0, \mu_2 \geq 0 \\ \mu_1 [(x_1+1)^2 + (x_2-1)^2 - 1] = 0 \\ \mu_2 [x_1^2 + (x_2-1)^2 - 1] = 0 \\ (x_1-1)^2 + (x_2-1)^2 \leq 1 \\ (x_1-1)^2 + x_2^2 \leq 1 \end{cases}$$

① g_1, g_2 都是 inactive. $\Rightarrow \mu_1^* = \mu_2^* = 0 \Rightarrow x^* = (0, 0)$, 矛盾!② g_1 active, g_2 inactive $\Rightarrow \mu_2^* = 0, x^* = (1 - \frac{\sqrt{2}}{2}, 1 - \frac{\sqrt{2}}{2})$ 为最小值!③ g_1 inactive, g_2 active $\Rightarrow \mu_1^* = 0, x_2^* = 0, x_1^* = 0$ 或 2 (矛盾!)④ 两者都工作, $x_2^* = \frac{1}{2}, x_1^* = 1 \pm \frac{\sqrt{3}}{2}$. 代入 stationary 条件, 有 $\mu_2 < 0$, 矛盾! $\Rightarrow$ 最小值为 $2(1 - \frac{\sqrt{2}}{2})^2 = 3 - 2\sqrt{2}$.

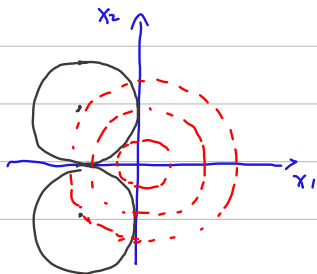
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5 Consider the following problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & f(x) = x_1^2 + x_2^2 \\ \text{s.t.} \quad & (x_1 + 1)^2 + (x_2 - 1)^2 \leq 1 \\ & (x_1 + 1)^2 + (x_2 + 1)^2 \leq 1 \end{aligned}$$

- (a). Sketch the feasible set and the level set of the objective function. Find the optimal point x^* and the optimal value f^* .
- (b). Write down the KKT conditions. Do there exist Lagrange multipliers that satisfy the KKT conditions? Is x^* regular?

(a)



feasible set: $(-1, 0)$

level set: 红线所示

$$x^* = (-1, 0), \quad f^* = (-1)^2 + 0^2 = 1.$$

$$(b) \quad (2x_1, 2x_2) + \mu_1 (2x_1 + 1, 2x_2 - 1) + \mu_2 (2x_1 + 1, 2x_2 + 1) = (0, 0) \quad \text{--- (1)}$$

$$\mu_1, \mu_2 \geq 0$$

$$(x_1 + 1)^2 + (x_2 + 1)^2 \leq 1 \quad \text{--- (2)}$$

$$(x_1 + 1)^2 + (x_2 - 1)^2 \leq 1 \quad \text{--- (3)}$$

$$\mu_1 [(x_1 + 1)^2 + (x_2 + 1)^2 - 1] = 0$$

$$\mu_2 [(x_1 + 1)^2 + (x_2 - 1)^2 - 1] = 0$$

由②③解出 $x_1 = -1, x_2 = 0$, 与①式矛盾!

故没有能满足KKT条件的解。

而 $\nabla g_1(x^*) = (0, -2)$, $\nabla g_2(x^*) = (0, 2)$, 是线性相关的,

故 x^* 不为正则点。

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6. Consider the following problem

$$\begin{aligned} \min \quad & (x_1 - \frac{3}{2})^2 + (x_2 - \frac{3}{4})^2 \\ \text{s.t.} \quad & \begin{cases} -x_1^2 + x_2 \geq 0 & \Rightarrow x_1^2 - x_2 \leq 0 \\ x_1 + x_2 \leq 4 \\ x_1, x_2 \geq 0 & \Rightarrow -x_1 \leq 0, -x_2 \leq 0 \end{cases} \end{aligned}$$

$$\begin{aligned} \text{KKT: } & \left(2x_1 - 3, 2x_2 - \frac{3}{2} \right) + \mu_1 (2x_1, -1) + \mu_2 (1, 1) - \mu_3 (1, 0) - \mu_4 (0, 1) = 0 \\ & x_1^2 - x_2 \leq 0, \quad x_1 + x_2 \leq 4, \quad -x_1 \leq 0, \quad -x_2 \leq 0 \\ & \mu_1 (x_1^2 - x_2) = 0 \\ & \mu_2 (x_1 + x_2 - 4) = 0 \\ & \mu_3 x_1 = 0 \quad \mu_4 x_2 = 0 \\ & \mu_i \geq 0 \end{aligned}$$

① $x_1 = \frac{3}{2}, x_2 = \frac{3}{4}$

$\Rightarrow \mu_3 = \mu_4 = 0, \mu_1 = 0, \mu_2 = 0$. 但不满足 $x_1^2 - x_2 \leq 0$, 不成立!

② $x_1 = 0, x_2 = 3$

$\Rightarrow \mu_4 = 0, \mu_1 = \mu_2 = 0$, 但不满足 stationarity 等式。

③ $x_1 = x_2 = 1$

$\Rightarrow \mu_3 = \mu_4 = 0, \mu_2 = 0$, 解得 $\mu_1 = \frac{1}{2}$, 满足 KKT.

且 $\nabla g_1(1, 1) = (2, -1)$, 线性无关

故 $x^* = x^{(3)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, f^* = \frac{5}{16}$.

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第七次

1. **Lasso.** Implement the projection onto ℓ_1 ball. Use projected gradient descent to solve the Lasso problem on slide 12 of §11. You should complete the implementation of projected gradient descent method in proj.gd.py.

$$\min_w \frac{1}{2} \|Xw - y\|_2^2 \quad X = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad y = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}, \quad t = 1$$

$$\text{s.t. } \|w\|_1 \leq t$$

Use the initial point $w_0 = (0, 1.5)$ and 0.1 step size. Report the solution and the number of iterations. Plot the trajectory of w_k and the gap $f(w_k) - f(w^*)$.

解法:
proj-gd: $x_{k+1} = P_X(x_k - t_k \nabla f(x_k))$.

$$P_X(y) = \arg \min_{x \in X} \|x - y\|_2^2$$

① 计算 $P_X(y)$:

$$\arg \min \|y - x\|_2^2$$

设 $P_X(x) = y$, 则 $\text{s.t. } \|y\|_1 \leq t$. 由对称性设 $y \geq 0$, $x \geq 0$, 则有

$$\min_x \frac{1}{2} \sum_{i=1}^n (x_i - y_i)^2 \quad \text{由 KKT, 存在 } \mu_i \geq 0, \text{ 使 } \forall i, (x_i - y_i) + \mu_i = 0$$

$$\text{s.t. } \begin{cases} \sum_{i=1}^n x_i \leq t \leftarrow \mu_0 \\ x_i \geq 0 \leftarrow \mu_i \end{cases} \Rightarrow x_i = y_i - \mu_0 - \mu_i = (y_i - \mu_0)^+$$

$$\therefore \|x\|_1 = \sum_{i=1}^n (y_i - \mu_0)^+ \leq t$$

- If $\|y\|_1 \leq t$, then $\mu_0 = 0$, $x = y$.
- If $\|y\|_1 > t$, then $\mu_0 > 0$ and x can be found in a similar way as in the water filling solution.
 - Sort y s.t. $y_1 \geq y_2 \geq \dots \geq y_n$.
 - Let

$$c_k = \frac{1}{k} \left(\sum_{i=1}^k y_i - t \right)$$

Then

$$\mu_0 = c_{k_0}$$

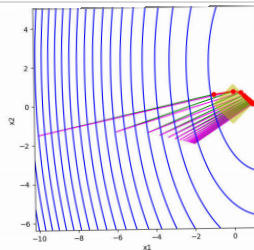
where

$$k_0 = \max\{k : c_k \leq y_k\}$$

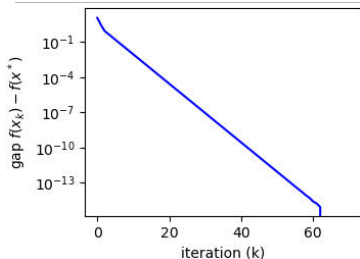
② 1) solution & iterations;

number of iterations: 70
solution: [9.99999998e-01 1.89170871e-09]
value: 4.5

2) trajectory:



3) gap:



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2. Let $x \in \mathbb{R}^3$. Consider

$$\begin{aligned} \min_x \quad & f(x) = e^{2x_1} + e^{x_2} + e^{x_3} \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 1 \end{aligned}$$

$$(a) F(x) = e^{2x_1} + e^{x_2} + e^{x_3} + \lambda (x_1 + x_2 + x_3 - 1)$$

求偏导得:
$$\begin{cases} 2e^{2x_1} + \lambda = 0 \\ e^{x_2} + \lambda = 0, e^{x_3} + \lambda = 0 \\ x_1 + x_2 + x_3 = 1 \end{cases} \Rightarrow \begin{cases} x_1 = \\ x_2 = x_3 = \frac{2+\ln 2}{5} \\ \lambda = -e^{\frac{2+\ln 2}{5}} \end{cases}$$

$$\therefore \lambda^* = \quad, \quad f^* =$$

(b)

$$A = (1, 1, 1)$$

$$\begin{bmatrix} \nabla^2 f & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ \lambda \end{bmatrix} = \begin{bmatrix} -\nabla f \\ 0 \end{bmatrix}$$

$$\begin{cases} \nabla^2 f v + A^T \lambda = -\nabla f & \text{--- ①} \\ A v = 0 & \text{--- ②} \end{cases}$$

$$\Rightarrow \text{由①有 } v = (\nabla^2 f)^{-1} (-\nabla f - A^T \lambda) = -(\nabla^2 f)^{-1} \nabla f - (\nabla^2 f)^{-1} A^T \lambda$$

$$\text{代入②有 } \lambda = -[A(\nabla^2 f)^{-1} A^T]^{-1} [A(\nabla^2 f)^{-1} \nabla f]$$

$$\therefore v = -(\nabla^2 f)^{-1} (\nabla f + A^T \lambda) = -(\nabla^2 f)^{-1} \nabla f - (\nabla^2 f)^{-1} A^T \lambda$$

(c)

```

(Damped) Newton's method with feasible start
1: initialization  $x \leftarrow x_0 \in X$   $\triangleright x_0$  is feasible, i.e.  $Ax_0 = b$ 
2: repeat
3:   Compute Newton's direction  $v$  by solving
      
$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ \lambda \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}$$

4:    $t \leftarrow 1$   $\triangleright$  backtracking line search on lines 4-7
5:   while  $f(x + tv) > f(x) + \alpha \nabla f(x)^T v$  do
6:      $t \leftarrow \beta t$ 
7:   end while
8:    $x \leftarrow x + tv$ 
9:   until  $\|v\| \leq \delta$ 
10:  return  $x$ 

Note. We cannot use  $\|\nabla f(x)\| \leq \delta$  as stopping criterion now, as

```

output:

```

iteration 0: [0. 1. 0.]
iteration 1: [-0.11369186 0.56845929 0.54523257]
iteration 2: [-0.09499115 0.5534111 0.54158005]
iteration 3: [-0.08601059 0.54598966 0.54002094]
iteration 4: [-0.0816069 0.5423022 0.53930469]
iteration 5: [-0.07942602 0.54046402 0.538962 ]
iteration 6: [-0.07834074 0.53954628 0.53879446]
iteration 7: [-0.07779938 0.53908775 0.53871163]
iteration 8: [-0.07752902 0.53885857 0.53867046]
iteration 9: [-0.07739392 0.53874399 0.53864993]
iteration 10: [-0.07732639 0.53868671 0.53863968]
iteration 11: [-0.07729263 0.53865807 0.53863455]
iteration 12: [-0.07727575 0.53864376 0.538632 ]
iteration 13: [-0.07726731 0.5386366 0.53863072]
iteration 14: [-0.07725887 0.53862944 0.53862944]
iteration 15: [-0.07725887 0.53862944 0.53862944]
optimal value: 4.28414144031119

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3. Consider the LP in standard form,

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

We are going to solve it using the barrier method.

$$(a) - \sum_{i=1}^n \log(x_i)$$

$$\therefore \text{为 } \min c^T x - \frac{1}{\epsilon} \sum_{i=1}^n \log(x_i) \triangleq g(x)$$

$$\text{s.t. } Ax = b$$

$$(b) \nabla g = c - \frac{1}{\epsilon} \cdot \frac{1}{x}$$

$$\nabla^2 g = \text{diag}\left(\frac{1}{\epsilon} \cdot \frac{1}{x^2}\right)$$

(c)

- 1: Choose a **strictly feasible** x_0 s.t. $Ax_0 = b$ and $g(x_0) < 0$, and $t_0 > 0$
- 2: **for** $k = 0, 1, 2, \dots$ **do**
- 3: $x_{k+1} \leftarrow \underset{x: Ax=b}{\operatorname{argmin}} \{f(x) + \frac{1}{t_k} B(x)\}$ **starting from** x_k
- 4: **if** $t_k \geq m/\epsilon$ **then**
- 5: **return** x_{k+1}
- 6: **end if**
- 7: $t_{k+1} \leftarrow \rho t_k$
- 8: **end for**

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & -x_1 - 3x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 8 \\ & -x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$$\text{slack: } 2x_1 + x_2 + x_3 = 8$$

$$-x_1 + x_2 + x_4 = 6$$

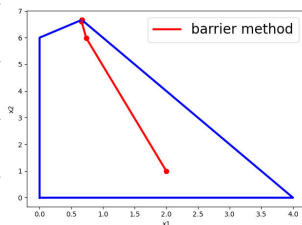
$$x_1, x_2, x_3, x_4 \geq 0$$

$$\therefore A = \begin{pmatrix} 2 & 1 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{pmatrix}, b = \begin{pmatrix} 8 \\ 6 \end{pmatrix}, c = (-1, -3, 0, 0)$$

输出结果:

```
iteration 0: [2. 1. 3. 7.]
iteration 1: [0.72784794 5.98281049 0.54209263 0.75423745]
iteration 2: [0.66315639 6.6016747 0.87201253 0.06148169]
iteration 3: [6.66182195e-01 6.66016785e+00 7.46775970e-03 6.01434613e-03]
iteration 4: [6.66616818e-01 6.66601667e+00 7.49690718e-04 6.00144040e-04]
iteration 5: [6.66616699e-01 6.66660167e+00 7.49892477e-05 5.99954704e-05]
iteration 6: [6.66662160e-01 6.66666081e+00 6.74796366e-06 5.39881560e-06]
iteration 7: [6.66666638e-01 6.66666630e+00 4.23748096e-07 3.37426130e-07]
iteration 8: [6.66666651e-01 6.66666726e+00 1.78014176e-07 1.42403585e-07]
iteration 9: [6.66666674e-01 6.66666744e+00 5.15394233e-08 4.12350459e-08]
optimal value: -20.6666690863538
```

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74.
$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & -x_1 - 3x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 8 \quad y_1 \\ & -x_1 + x_2 \leq 6 \quad y_2 \\ & x_1, x_2 \geq 0 \quad y_3, y_4 \end{aligned}$$

(a) $(-2y_1 + y_2 + y_3)x_1 + (y_1 - y_2 + y_4)x_2 = -x_1 - 3x_2 \geq -8y_1 - 6y_2$

则 $\max -8y_1 - 6y_2$

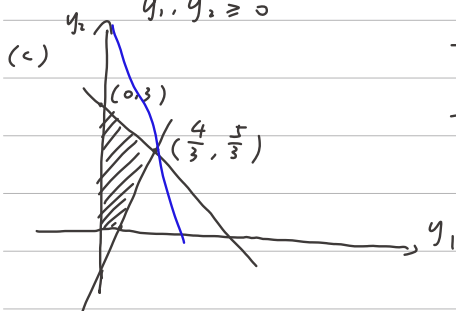
s.t.
$$\begin{cases} 2y_1 - y_2 - y_3 = 1 \\ y_1 + y_2 - y_4 = 3 \end{cases}$$

$y_1, y_2, y_3, y_4 \geq 0$

(b) symmetrical dual LP:

$\max -8y_1 - 6y_2$

s.t.
$$\begin{cases} 2y_1 - y_2 \leq 1 \\ y_1 + y_2 \leq 3 \\ y_1, y_2 \geq 0 \end{cases}$$



取 $y_1 = \frac{4}{3}, y_2 = \frac{5}{3}$, 有

$-8y_1 - 6y_2 = -\frac{62}{3} = -20.667,$

与 3(d) 结果相同。

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